

rFindMaxAr

Write a **recursive** C function that finds the maximum number in an array of integer numbers. In the function, the parameter `ar` accepts an array passed in from the calling function. The integer parameter `size` indicates the size of the array. The pointer parameter `max` is used for passing the maximum number to the caller via call by reference. The function prototype is given as follows:

```
void rFindMaxAr(int *ar, int size, int *max);
```

A sample program template is given below to test the function:

```
#include <stdio.h>
void rFindMaxAr(int *a, int size, int *max);
int main()
{
    int ar[50], i, max, size;

    printf("Enter array size: \n");
    scanf("%d", &size);
    printf("Enter %d numbers: \n", size);
    for (i=0; i < size; i++)
        scanf("%d", &ar[i]);
    max=ar[0];
    rFindMaxAr(ar, size, &max);
    printf("rFindMaxAr(): %d\n", max);
    return 0;
}
void rFindMaxAr(int *ar, int size, int *max)
{
    /* Write your code here */
}
```

Some sample input and output sessions are given below:

(1) Test Case 1:

```
Enter array size:
5
Enter 5 numbers:
1 2 3 4 5
rFindMaxAr(): 5
```

(2) Test Case 2:

```
Enter array size:
7
Enter 7 numbers:
2 5 4 -7 9 10 1
rFindMaxAr(): 10
```

(3) Test Case 3:

```
Enter array size:  
3  
Enter 3 numbers:  
-1 -3 -2  
rFindMaxAr(): -1
```